Helpyard Project Report

Architecture, dashboard sections, bridge routes, Windows actions, safeguards, testing, and release readiness

REPORT	REVIEW DATE	PROJECT ROOT
02	17 October 2026	O:\Projects\mcp\MCP

Assessment at a glance
Helpyard is a local-first Windows workstation dashboard with a browser UI and a Python standard-library bridge. The default setup has no pip or npm install step. Core protections include loopback binding, one-time browser bootstrap, in-memory bridge credentials, input checks, and approval gates. It remains a prototype: release qualification and persistent audit history are not complete.

Scope and evidence

This report synthesizes the readable project source, launcher, tests, VS Code configuration, README, and resource documentation. The review examined the Python bridge, legacy PowerShell bridge, dashboard markup and inline application logic, the 12-section navigation, available API routes, and all named Windows actions. Existing binary PDFs and generated Python bytecode were not inspected. No tests or live system-changing actions were run as part of producing this report; references to earlier documented test results are identified as project documentation, not independently repeated verification.

Project inventory

Path	Role
README.md	Setup, behavior, security boundaries, troubleshooting, tests, and documented limits.
Start-Helpyard.bat	Windows one-click launcher. Prefers Python launcher, checks Python 3.10+, starts bridge.py minimized and opens browser.
bridge.py	Default local HTTP API/server, authentication/bootstrap, Windows queries/actions, Ollama proxy, and instance lock.
bridge.ps1	Legacy/manual PowerShell server implementing overlapping API and system operations.
pc-control-dashboard.html	Single-page UI, styling, charting, navigation, settings, local/cloud chat, and request handling.
tests/test_bridge.py	Python unittest coverage for bridge HTTP behavior and guards.
tests/dashboard-check.js	Static JavaScript/HTML structural checks and secret-storage/API correspondence assertions.
tests/bridge-smoke.ps1	PowerShell smoke checks for the legacy bridge.
.vscode/launch.json; .vscode/tasks.json	Static dashboard preview, legacy bridge launch, and default Python bridge task.
resources/*.md; resources/*.pdf	Resource catalog, modernization plan, implementation strategy, and existing PDF deliverables.

The source documentation identifies Python 3.10+ and Windows PowerShell 5.1 for Windows operations. The default Python server uses the standard library; the dashboard uses native HTML/CSS/JavaScript/canvas. Ollama is an optional separately

installed service; cloud chat uses user-provided provider credentials.

1. Architecture and startup

- The batch launcher changes to the project directory, verifies bridge.py, detects py/python availability and Python version, then starts bridge.py with --open-browser.
- bridge.py binds to 127.0.0.1, tries configured candidate ports beginning at 8787, and uses a Windows named mutex to avoid multiple instances for the same folder.
- The dashboard is served at /pc-control-dashboard.html. A 60-second, one-use connection code is placed in the URL fragment. The page removes it from browser history, exchanges it for a bearer token, and keeps the token in JavaScript memory.
- The server accepts localhost clients and checks Host and Origin, uses authenticated API routes, bounds JSON bodies to 1 MiB, and caps concurrent request workers at 16.
- Fixed PowerShell scripts perform reviewed Windows operations; browser values are validated, and service names are quoted before insertion into the fixed PowerShell operation.
- Ollama is restricted to a configured localhost HTTP origin. Project discovery scans this project and immediate child Git repositories, not arbitrary paths.
- Cloud AI calls go directly from the browser to the chosen provider. Cloud prompts leave the device; the API key is intended for the current session only.

Data persistence: theme, accent, tip visibility, and selected non-secret preferences are saved in browser local storage. Bridge tokens and cloud keys are excluded from the persisted-state shape. Overview chart samples are temporary in-memory data. The Security page is descriptive and does not represent a durable audit trail.

2. Dashboard sections

Section	Function and boundary
Overview	CPU activity, used memory, uptime, process count, processor topology, fixed-disk capacity, recent chart, Ollama snapshot, and shortcuts. Samples are 5 seconds apart, max 24, only while Overview is open; no long-term history.
Projects	Shows project root and immediate child Git repositories with branch/status discovery. Not a general disk crawler.
Workbench	Saves prompt guidance and displays a limited set of named actions. Instructions do not authorize tools or change bridge policy.
Processes	Lists up to 60 processes ordered by memory, with filters and stop action. Per-process CPU figure is cumulative seconds, not utilization.
Services	Lists Windows services with filters and start/stop/restart controls. Mutations require confirmation and explicit API approval.
Terminal	Shell input and Run control are disabled. Arbitrary /api/run is denied by the bridge.
Local AI	Checks Ollama and installed models; supports local chat through the bridge. Distinguishes server unavailable from no installed models.
Cloud AI	Supports Anthropic, OpenAI, and Gemini provider choices. Requests are direct to provider; sensitive prompts should not be sent.
MCP	Status display only; startup intentionally disabled pending reviewed server registry and fixed executable/argument policy.
Skills	Read-only guidance cards and navigation shortcuts; no downloading, installation, or execution.
Security	Displays safeguard posture and descriptive audit entries. It is not persistent event logging.
Settings	Appearance choices, contextual tips, bridge URL/token manual connect, and reset saved browser data.

3. HTTP/API inventory

The Python bridge routes observed in source are summarized below. Requests other than the dashboard/probe/bootstrap require the in-memory bearer token. The legacy PowerShell bridge contains overlapping routes but differs in authentication/bootstrap and implementation details; it is not the launcher default.

Method and route	Purpose / access control
GET /, /pc-control-dashboard.html	Serves the dashboard without API token; request must pass local Host/Origin checks.
GET /api/bridge-probe	Unauthenticated local origin probe, HTTP 204; used for bridge URL discovery.
OPTIONS (API)	Local CORS preflight; advertises Authorization, Content-Type and GET/POST/OPTIONS.
POST /api/bootstrap	One-time code exchange. Requires exact 127.0.0.1 origin, valid unexpired code; replay denied.
GET /api/ping	Authenticated basic bridge, admin, host, and user status.
GET /api/health	Authenticated Windows OS/build and uptime information.
GET /api/actions	Authenticated action registry with risk, permission and approval metadata.
GET /api/projects	Authenticated workspace/immediate child Git repository list.
GET /api/stats	Authenticated CPU, RAM, processor topology, disks, uptime and process count.
GET /api/processes	Authenticated process inventory (up to 60 items).
GET /api/services	Authenticated Windows service inventory.
GET /api/mcp/status	Returns an empty list in the default Python bridge; no MCP lifecycle is wired.
GET /api/ollama/tags; POST /api/ollama/chat	Authenticated Ollama model listing and non-streaming chat via configured localhost service.
GET /api/claude-config	Reads current-user Claude Desktop configuration path/content.
POST /api/process/stop	Approval-gated process termination; rejects invalid IDs and self-termination.
POST /api/service	Approval-gated named service start/stop/restart.
POST /api/action	Executes a registered action only; elevated rights and approval checked per action.
POST /api/run	Always denied: arbitrary PowerShell execution disabled.
POST /api/mcp/start	Always denied until reviewed registry is implemented.
POST /api/mcp/stop	Default Python bridge returns success without lifecycle behavior; no server is started by current Python bridge.
POST /api/claude-config	Approval-gated JSON config update with timestamped backup when prior file exists.

Errors generally use explicit HTTP status codes and JSON messages. Python bridge unexpected errors are logged to its console and return a generic server error to the browser.

4. Registered Windows actions

The Python registry contains 16 named actions. ?Approval? below means explicit confirmation represented in the API request; it is not a separate OS privilege boundary. Some actions also require administrator rights.

Action	Effect	Risk / approval / privilege
lock	Locks current Windows session.	Low / no approval / standard
sleep	Suspends computer.	Medium / required / standard

Action	Effect	Risk / approval / privilege
restart	Schedules restart after 30 seconds.	High / required / admin
shutdown	Schedules shutdown after 30 seconds.	High / required / admin
cancel	Cancels pending shutdown/restart.	Low / no approval / standard
flushdns	Flushes Windows DNS cache.	Low / no approval / standard
taskmgr	Opens Task Manager.	Low / no approval / standard
explorer	Opens File Explorer.	Low / no approval / standard
settings	Opens Windows Settings.	Low / no approval / standard
devmgr	Opens Device Manager.	Low / no approval / standard
emptybin	Empties Recycle Bin.	Medium / required / standard
cleantemp	Recursively removes current user TEMP contents.	Medium / required / standard
perf_high	Activates high-performance power scheme.	Medium / required / admin
perf_balanced	Activates balanced power scheme.	Low / no approval / standard
perf_saver	Activates power-saver scheme.	Low / no approval / standard
winupdate	Opens Windows Update settings.	Low / no approval / standard

Operational caution: approval does not eliminate the impact of a mistake. Process termination may lose unsaved work; service changes may affect Windows; temp/recycle-bin cleanup deletes data; sleep/restart/shutdown affect availability. Verify targets before use.

5. Security, privacy, and reliability review

Area	Observed strengths	Residual considerations
Network exposure	Python server binds IPv4 loopback; validates remote address, Host and Origin; authenticated API; no-store responses; one-use bootstrap.	Keep port private. Validate default and legacy implementation behavior separately; do not treat browser confirmation as least privilege.
Credentials	Cryptographic URL bootstrap code, expiry/replay defense, constant-time token compare; token held in memory; cloud key not saved.	A pasted manual token is still a bearer credential. Cloud prompts go to providers. Avoid exposing console/connection URL or sensitive prompt contents.
Input / execution	1 MiB body bound; object-only JSON; allowlisted fields; typed approval; exact service-name pattern; fixed scripts; no arbitrary shell/MCP start.	Review each future action and argument path. Named destructive actions remain consequential. Keep disabled features disabled until a registry and policy are reviewed.
Local proxy	Ollama URL normalized to localhost HTTP; response bounded to 1 MiB; chat messages validated; 600-second chat timeout.	HTTP is acceptable only for the local-only configuration; Ollama is optional and not security-isolated from local processes.
UI rendering	Dynamic text is generally escaped or placed using textContent; cloud key input is password type and cleared on initialization.	Client-side checks do not replace bridge enforcement. Perform full accessibility, browser, injection and privacy review before release.
Audit and resilience	Bounded server workers; request timeouts in frontend; explicit route failures; timestamped Claude config backup.	Persistent audit history is absent. Some long-running Windows and Ollama calls need user-facing operational verification. Legacy PowerShell implementation creates a second maintenance/security surface.

Report observations: the Python bridge is the actual default runtime, while bridge.ps1 is separately maintained and has different behavior. The Python GET /api/mcp/status is a fixed empty result; POST /api/mcp/stop is a no-op success response. These responses should not be interpreted as a real MCP process manager. The Security panel is static summary content rather than an audit service.

6. Tests and verification

Check	Documented scope	This report run
Python unittest	Dashboard serving; bootstrap origin/expiry/replay; token auth; data routes; approvals; malformed/oversized JSON; slow Ollama concurrency; CPU topology query. Source lists 7 test methods.	Not run. README documents command: py -3 -m unittest discover -s tests -p test_bridge.py -v.
Dashboard JavaScript check	Parses inline script; validates page/nav parity, contextual tips, DOM IDs, route mapping, no external assets, disabled terminal, secret storage safeguards and processor fields.	Not run. README documents: node .\tests\dashboard-check.js.
Legacy PowerShell smoke	Tests only bridge.ps1, including auth/origin/host handling, metrics, approval gates, disabled operations and input validation.	Not run. README documents PowerShell command in project root.
Live OS / provider behavior	Python unit tests stub Windows/Ollama behavior; live actions and provider calls require separate environment verification.	Not performed; no system-changing actions or external provider calls were made.

Suggested validation sequence

- Run all three project-documented test commands from the repository root; capture outputs and investigate failures.
- On supported Windows versions, verify launcher startup, selected port, fresh-browser one-time bootstrap, bridge reconnect and real read-only metrics.
- Test Ollama unavailable, available/no models, model selection, and local chat using a non-sensitive prompt.
- Review every mutation with a disposable process/service or safe test target; never validate shutdown, restart, cleanup, or other destructive actions on a production workstation.
- Check keyboard-only use, focus order, reduced motion, contrast, screen-reader labels, zoom, and narrow viewport; record browser/OS versions.
- Verify browser storage does not contain session token or provider key, and assess how provider calls and logs handle sensitive information.

7. Prioritized recommendations

Priority	Recommendation	Reason / acceptance evidence
P1 - before release	Add a real, durable, privacy-conscious audit trail for approved mutations, or remove/rename UI language that implies an audit log.	Record time, action, target, approval, outcome without logging secrets; document retention and test failure handling.
P1 - before release	Perform a focused security review and mutation-safety review of both bridge implementations; decide whether to keep or retire the legacy bridge.	Document supported path and verify equivalent auth, validation, error, backup, and approval semantics.
P1 - before release	Complete Windows/browser qualification and accessibility evaluation.	Fresh-profile bootstrap, metrics, narrow layouts, keyboard/screen reader, contrast/zoom evidence across supported platforms.
P2 - product decision	Clarify MCP status/stop behavior and remove success-shaped no-op routes or implement real, reviewed semantics.	API/UI must clearly state unsupported behavior; test route result and documented capability match.
P2 - product decision	Review whether process/service/action controls should be opt-in or more tightly scoped.	Least-privilege rationale, explicit target display, confirmation, and test coverage for each mutation.
P3 - ongoing	Preserve zero-dependency/offline policy unless a measured need justifies a reviewed dependency.	Source/version/license/integrity and offline behavior documented; no automatic skill/MCP execution.

8. Overall assessment

Conclusion: Helpyard has a clear local-prototype architecture and several meaningful guardrails. The default is intentionally constrained and dependency-light. The main release gaps are qualification rather than absent core structure: no persistent audit history, incomplete Windows/browser/accessibility evidence, an additional legacy bridge to maintain, and placeholder MCP lifecycle responses. Do not describe this snapshot as release-ready without the documented checks and follow-up work.

Referenced project documentation

README.md; resources/RESOURCE-CATALOG.md; resources/MODERNIZATION-PLAN.md; resources/Helpyard-Implementation-Strategy.md. Existing binary PDF contents were not assessed while preparing this report. This report is a source-based engineering overview, not a formal penetration test or certification.